



Improving school readiness of high-risk preschoolers: Combining high quality instructional strategies with responsive training for teachers and parents



Susan H. Landry^{a,*}, Tricia A. Zucker^a, Jeffrey M. Williams^a, Emily C. Merz^b,
Cathy L. Guttentag^a, Heather B. Taylor^a

^a The University of Texas Health Science Center at Houston, Children's Learning Institute, 7000 Fannin Street, Suite 2300, Houston, TX 77030, USA

^b Columbia University, Department of Epidemiology, 722 West 168th Street, 720F, New York, NY 10032, USA

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ABSTRACT

This study evaluated whether the combination of two proven interventions, one in Head Start classrooms (The Early Education Model, TEEM) and one in the home (Play and Learning Strategies, PALS) resulted in enhanced effects on at-risk 3- to 5-year-old children's school readiness skills when compared to either of these interventions alone. Teachers and parents were trained to use a responsive style and strategies that supported children's school readiness skills with the goal of providing children consistency in responsive practices across the school and home environments. The study was conducted in 77 classrooms with teachers randomized to either the TEEM ($n = 39$) or No TEEM (i.e., control or business as usual, $n = 38$) conditions. Six to eight children in each classroom were randomly assigned to either have their parents receive PALS ($n = 314$; 210 after attrition) or to a No PALS condition ($n = 309$; 221 after attrition) resulting in four conditions: TEEM/PALS, TEEM/No PALS, No TEEM/PALS, and No TEEM/No PALS. Results showed greater gains in the TEEM teachers' language and literacy instructional practices and sensitivity compared to control teachers, but there were few significant findings for child cognitive outcomes. Parents receiving PALS, as compared to those without PALS, showed greater increases across play and book reading contexts in numerous responsive behaviors linked to the attachment and socio-cultural theories. Children whose parents received PALS versus those whose parents did not showed greater gains in direct measures of print knowledge and self-regulation and in social and language skills observed during interactions with their parent. Interactive effects of TEEM plus PALS were seen for increased engagement in shared book reading but not for other cognitive or social outcomes.

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1. Introduction

There is growing consensus that providing young children, particularly those from low socioeconomic status (SES) homes, with high-quality classroom experiences prior to entering kindergarten is a key to closing a well-documented achievement gap (e.g., National Assessment of Educational Progress, 2009). High quality prekindergarten (pre-k) experiences increase the likelihood that children will enter school with the skills necessary for academic success (e.g., Heckman, Moon, Pinto, Savelyev, & Yavitz, 2010). However, many existing pre-k programs are not of high enough quality to assure that children from higher risk backgrounds will

get the learning experiences they need to be school ready (e.g., Burchinal, Kainz, & Cai, 2011). For example, in the most recent impact study of Head Start, a federally funded preschool program for children from very low-SES homes, only a few small positive effects in cognitive and social skills occurred across pre-k, and these did not persist through first grade (USDHHS & ACF, 2010). Also, interventions implemented to enhance the effectiveness of pre-k programs, like Head Start, have yielded mixed results, and when there are improvements they are often small (Burchinal et al., 2011).

One potential solution to this problem of small impacts for many pre-k programs serving low-SES students is to increase parent responsiveness and support for school readiness in children's home environments. Many pre-k programs, particularly Head Start, acknowledge the importance of having a parent component (Zigler & Muenchow, 1992), but parent involvement varies greatly across

* Corresponding author.

E-mail address: Susan.Landry@uth.tmc.edu (S.H. Landry).

Head Start programs due, in part, to the quality of parenting programs chosen (Cooper & Lanza, 2014). There are few examples of Head Start programs incorporating parent programs that have evidence of effectiveness from experimental studies (Cooper & Lanza, 2014). There is a dearth of experimental studies evaluating effects on young children's school readiness of a combined approach in which a high-quality pre-k classroom program is paired with an evidence-based parent program. This study examined the individual and combined impact of a school and home intervention that have both been shown to be effective.

1.1. Rationale for intervening in the home and classroom

Children from low-SES backgrounds often enter preschool behind in school readiness skills (e.g., Son & Morrison, 2010). School readiness is a multidimensional construct including cognitive skills (e.g., oral language, early literacy), social and behavioral skills, and self-regulation skills. The learning gains required for children from high-risk homes to be prepared for school are often not realized and can ultimately contribute to high dropout rates (Duncan & Sojourner, 2013) or reduced benefits from schooling (Lynch, Law, Brinkman, Chittleborough, & Sawyer, 2010). Early childhood teachers' ability to provide the quality and intensity of learning experiences necessary for children from low-SES backgrounds to "catch up" in their development is hindered by many factors, one of which is teachers' limited professional preparation and knowledge (Burns, Donovan, & Bowman, 2000). Educating young children requires teachers to have a deep understanding of curriculum content, pay attention to individual learning needs, and reflect on their teaching practices (e.g., Burns et al., 2000).

Research demonstrates that caregivers in the home environment can facilitate young children's development of cognitive and social skills so they are better prepared to enter school (e.g., Bakermans-Kranenburg, Van IJzendoorn, & Juffer, 2003). However, this is less likely to happen in homes where parents have limited education and low incomes. Multiple life stresses, poor parental role models, and a limited understanding of their important role as their child's teacher are contributing factors to this cycle (e.g., McLoyd & Wilson, 1990). Poverty is associated with low levels of rich language input (Hart & Risley, 1995), decreased experiences with books (Federal Interagency Forum on Child & Family Statistics, 2009), less responsiveness to children's signals and needs (Landry, Smith, Swank, Assel, & Vellet, 2001), and more frequent use of power-assertive techniques (McLoyd, 1990; Van Zeijl et al., 2006). Thus, children from low-SES backgrounds are less likely to experience parenting behaviors known to predict more optimal school outcomes (e.g., Wakschlag & Hans, 1999).

Given the pragmatic challenges of preparing children from low-SES backgrounds for school and research demonstrating the importance of responsive adults in supporting children's development, intervening in both the classroom and home environments may best promote children's school readiness. Theoretical frameworks that consider person-environment fit (Bronfenbrenner & Morris, 2006) and correlational research (Barbarin, Downer, Odem, & Head, 2010) suggest that providing pre-k children with congruent teacher and parent emotionally responsive, cognitively stimulating practices optimally supports school readiness. In turn, better alignment between these two environments may allow children to use their cognitive and emotional resources to learn more from their interactions with more competent adults. We further detail theoretical and empirical support for this theory of change below.

1.2. The value of responsive parenting practices

Parenting research across different theories and research frameworks (e.g., attachment: Ainsworth, Blehar, Waters, & Wall,

1978; socio-cultural: Rogoff, 1990) describes the importance of responsive adults for children's optimal early development. Responsiveness encompasses a range of emotionally and cognitively responsive adult behaviors and practices that collectively support children's social, behavioral, self-regulation and cognitive skills (e.g., Bornstein & Tamis-LeMonda, 1989; Grusec & Goodnow, 1994). Emotionally responsive parenting includes high levels of warm, reciprocal behaviors that acknowledge children's signals and needs in ways that are prompt and contingent to what children are signaling (e.g., Dunst & Kassow, 2008). Cognitively responsive parenting integrates rich language input and actions that build on and scaffold children's interests so children are appropriately challenged and supported as they learn new concepts and skills (e.g., Tamis-LeMonda, Bornstein, & Baumwell, 2001). Following the child's lead and responding contingently are thought to support learning, in part, by facilitating the child's development of mechanisms for coping with stress and novelty in his or her environment (Ainsworth et al., 1978). With repeated experiences of these responsive practices, trust is formed between the caregiver and child that allows the child to internalize this sense of security and generalize learning to new experiences. Evidence from experimental responsive parenting interventions demonstrates that when responsive practices are increased, children show gains in independent problem-solving, language, and social skills (Bakermans-Kranenburg et al., 2003; Landry, Smith, Swank, & Guttentag, 2008) as well as behavioral development (Van Zeijl et al., 2006).

1.3. The potential value of combined parent and teacher interventions

A central hypothesis of the current study was that children would benefit from the interactive effect of providing both high quality, responsive instruction in preschool and a responsive parenting intervention. There are several reasons for this expectation. As teachers are learning to implement organizational strategies, activities, and responsive interactions with children that promote language development, pre-literacy skills, and self-regulation, parents are simultaneously learning related skills at home, including how to promote their children's language skills, increase children's focused attention, manage behavior using positive strategies, and increase interactive book-reading. Thus, children may be receiving more exposure to activities that promote language, literacy, and attention-focusing skills than they would if only the home or school environment were receiving intervention (i.e., additional dosage). A second reason to expect an enhanced effect of this dual intervention model is related to increased alignment between home and school expectations and practices, which provides greater consistency for children. Churchill (2003) examined the concept of "goodness of fit" between the alignment of Head Start parents' and teachers' expectations of children and parenting behaviors, and found that the fit between parent and teacher was significantly positively correlated with children's observed social skills.

Few studies have experimentally examined the effects of parent-plus-teacher interventions on school readiness skills. A few have specifically examined the effects of dialogic reading interventions, programs that train adults to do shared book reading to support children's language development (see Mol, Bus, de Jong, & Smeets, 2008). Dialogic reading studies have produced mixed results when examining a parent-plus-teacher approach. One study reported better language outcomes for classroom only and classroom-plus-home interventions relative to a control group (Whitehurst et al., 1994), while another study found effects were largest for a home only condition, not a combination of a home and classroom intervention (Lonigan & Whitehurst, 1998). Yet another study examining a print referencing style of shared reading found

no added benefit of a school-plus-home intervention relative to classroom only print-focused condition (Justice, Logan, Kaderavek, & Dynia, 2015). Differences in outcomes across these studies may be due to a number of factors including low levels of implementation by parents in some studies and differences in program intensity. As all of these studies incorporated parent programs that specifically target book reading, an activity that is not always an everyday activity for low SES families, it raises the question of whether parent interventions that go beyond reading would show an added benefit when combined with a classroom based program.

There may be advantages for the implementation of more global parent responsiveness interventions versus one specifically targeting one home activity (e.g., shared book reading). A recent study in Head Start programs combined a classroom enrichment program with a parent program that extends from preschool through the transition to kindergarten (Bierman, Welsh, Heinrichs, Nix, & Mathis, 2015). Home visits trained parents to use responsive interactions, dialogic reading and various language and literacy activities. Relative to a comparison group, effects were found for literacy skills and teacher-report of academic and social skills but not for vocabulary or reading fluency. Other parent programs that target a broad range of responsive parenting behaviors through home-based coaching have been shown to impact children's learning in multiple contexts (e.g., everyday activities, play, book reading, independent play, and interactions with unfamiliar adults) and their language, social, and emotional skills (Guttentag et al., 2014; Landry et al., 2008; Van Zeijl et al., 2006). The addition of a coach working with families to support the use of new behaviors across everyday activities may have the added benefit of enhancing parent implementation compared with programs that target one specific activity (e.g., book reading). Results across this limited body of experimental research demonstrate the need for further examination of the added benefit of engaging parents of children from low-income homes to use responsive behaviors that support school readiness.

1.3.1. Parental and teacher support of children's skills

Parent and teacher support of several aspects of young children's development are of interest in this study because they are all required for school readiness. These domains include cognitive skills, social and behavioral skills, and self-regulatory processes. Cognitive skills include oral language and early literacy (e.g., letter knowledge, phonological awareness) skills, while social and behavior skills include behaviors such as cooperation, social engagement, and comfort with new situations. Self-regulation skills involve children's ability to sustain attention to goal-directed activity, resist interference from competing sources of stimulation, regulate their emotions, and respond effectively to situations that are motivationally significant (Garon, Bryson, & Smith, 2008). Children from low-SES homes often show difficulties developing these skills (e.g., Noble, Norman, & Farah, 2005).

Adults can support preschool children's cognitive skill development by providing a stimulating environment with engaging, playful learning activities (Burns et al., 2000). Caregivers and teachers can foster social, behavioral and self-regulation skills by providing sensitive support for children's immature attentional and communication skills, help in identifying the goals of a task, and in planning how to achieve the goal (e.g., Bernier, Carlson, Deschenes, & Matte-Gagne, 2012). Responsive parenting is related to the development of social and self-regulation skills, including tasks that require inhibiting a response when problems are emotionally arousing (i.e., effortful control: Bernier et al., 2012; Sroufe, Coffino, & Carlson, 2010). Several studies show positive effects on self-regulatory skills using interventions (e.g., curriculum, class-

room routines) that trained educators to support these skills (e.g., Domitrovich, Cortes, & Greenberg, 2007; Raver et al., 2011).

1.4. Components of aligned, evidence-based home and school interventions

The current study examined the separate and interactive benefits of classroom and parent interventions for preschool-age children in Head Start programs. The study design addressed a question that has not received much attention from experimental studies, namely the extent to which training adults (teachers, parents) across home and school settings to use similar responsive practices, but different types of learning activities, can promote school readiness skills. The interventions targeted key aspects of the home and classroom with the greatest potential to support preschool children's learning, namely responsive, intentional adult caregiving behaviors and language- and literacy-promoting activities (Hart & Risley, 1995; Pianta & Steinberg, 1992).

Experimental studies of the present parent intervention, Play and Learning Strategies (PALS), have shown significant increases in parents' use of behaviors that are responsive to children's signals and those that support children's language skills (range of effect sizes, $d = 0.30$ – 0.52). Increases in these behaviors, in turn, mediated significant increases (d range = 0.16 – 0.38) in children's language, cognitive and social-emotional skills (Landry et al., 2008; Landry, Smith et al., 2011). The framework for PALS includes four specific areas of responsiveness in the home environment documented to be important for promoting growth in cognitive and social skills: (1) maintaining vs. redirecting children's attentional focus and interests, (2) contingent responsiveness, (3) rich language input, and (4) warm sensitivity (Landry et al., 2008).

Within the classroom these same responsive practices were targeted and combined with training in the use of: (1) progress monitoring assessments, (2) whole- and small-group instruction for cognitive activities, (3) repeated exposure to concepts to expand understanding, and (4) a balance of teacher- vs. child-directed activities. The classroom intervention, The Early Education Model (TEEM), was implemented through face-to-face, small-group teacher trainings that used a web-based, multi-course program combined with in-classroom coaching to support implementation. In two large experimental studies, TEEM has demonstrated greater increases (moderate to large effect sizes) in teachers' practices compared to teachers who did not receive TEEM, as well as gains in children's language and literacy skills (small effect sizes) compared to children in control classrooms (Landry, Anthony, Swank, Monseque-Bailey, 2009; Landry, Swank, Anthony, & Assel, 2011).

1.5. Research questions

Using a randomized control design in which intervention conditions were completely crossed (i.e., PALS/No TEEM, TEEM/No PALS, PALS + TEEM, and No PALS/No TEEM), we considered three research questions. First, what are the impacts of the TEEM intervention on teacher behaviors? Second, what are the impacts of the PALS intervention on parent behaviors? We examined adult behaviors that constitute responsive practices and promote school readiness. Third, what are the benefits of each of the interventions on the school readiness skills of preschoolers from socioeconomically disadvantaged families? As part of this question we also asked, to what extent are additive or interactive effects present in child outcomes when the two interventions are implemented simultaneously?

In the home setting we provided the program in English or Spanish according to parents' preference. We hypothesized that the PALS intervention would promote changes in parent behaviors such as warm responsiveness and cognitive scaffolding, and

Table 1
Teacher/classroom, parent, and child characteristics by intervention group.

Variable	Teacher/Classroom		Parent			Child					Control	Total
	TEEM	No TEEM	Total	PALS	No PALS	Total	TEEM/PALS	TEEM/no PALS	PALS/no TEEM			
	<i>n</i> = 39	<i>n</i> = 38	<i>n</i> = 77	<i>n</i> = 210	<i>n</i> = 221	<i>n</i> = 431	<i>n</i> = 102	<i>n</i> = 112	<i>n</i> = 109			
Age in years <i>M</i> (<i>SD</i>)	–	–	–	–	–	–	4.42 (0.46)	4.38 (0.54)	4.34 (0.50)	4.40 (0.50)	4.38 (0.50)	
Female (%)	94.87	94.59	94.74	98.57	98.17	98.36	47.57	51.89	52.43	48.11	48.60	
Race/Ethnicity (%)												
African American	48.72	64.86	56.58	23.56	27.85	25.76	24.75	31.82	30.27	29.09	29.07	
Hispanic/Latino	33.33	21.62	27.63	72.12	67.58	69.79	74.26	64.55	68.81	70.91	69.53	
Caucasian/White	12.82	10.81	11.84	1.92	3.20	2.58	0.99	3.64	0.00	0.00	1.16	
Asian	2.56	2.70	2.63	1.44	0	0.70	0	0	0	0	0	
Years of teaching experience <i>M</i> (<i>SD</i>)	11.29 (9.03)	10.65 (9.11)	–	–	–	–	–	–	–	–	–	
Highest level of education (%)												
<High school diploma	0	0	0	36.24	31.51	33.80	–	–	–	–	–	
High school/GED	10.26	0.00	5.26	24.15	28.31	26.29	–	–	–	–	–	
CDA credential	28.21	18.92	23.68	–	–	–	–	–	–	–	–	
Some college	–	–	–	27.54	27.85	27.70	–	–	–	–	–	
Associate's degree	12.82	16.22	14.47	3.86	3.20	3.52	–	–	–	–	–	
Bachelor's degree	41.03	45.95	43.42	5.31	5.48	5.40	–	–	–	–	–	
Some graduate school	7.69	18.99	13.15	0.97	1.83	1.40	–	–	–	–	–	
Classroom or home language (%)												
Mostly English	66.66	75.67	71.05	–	–	–	39.22	49.11	40.74	46.85	44.01	
Both English and Spanish	28.21	16.22	22.37	–	–	–	20.59	15.18	16.67	14.41	16.59	
Mostly Spanish	5.13	8.11	6.58	–	–	–	40.20	35.71	42.59	38.74	39.17	
Free/reduced lunch program (%)	–	–	–	–	–	–	96.04	96.40	96.33	98.18	96.08	

Note. TEEM, The Early Education Model; PALS, Play and Learning Strategies; Control group =, no TEEM/no PALS group.

positive changes in children's school readiness, compared to the No PALS condition (Landry et al., 2008; Landry, Smith et al., 2011; Landry, Swank et al., 2011). For the TEEM classrooms, instruction and coaching on responsiveness and effective instructional approaches were expected to result in greater improvement in teachers' behaviors and children's school readiness outcomes compared to control classrooms (Landry, Anthony et al., 2009; Landry, Swank et al., 2009). We expected the greatest impact of combined parenting and classroom interventions. These changes in children's experiences, across two settings, were hypothesized to intersect to promote greater improvements in school readiness skills compared to each intervention alone or control condition.

2. Method

2.1. Research design

We recruited Head Start centers in the Houston and Austin metro areas across three academic years. Directors of six Head Start agencies were contacted and all agreed to participate. One classroom per building was eligible to participate each year. Classroom teachers were randomized to either the TEEM ($n = 39$) or 'No TEEM' control (i.e., business as usual, $n = 38$) conditions. Parents of children in all classrooms were invited to participate and sign informed consents. Of the 828 total children with parental consent, six to eight children per classroom were randomly selected for participation. Half of these children were randomly assigned to have their parents receive the PALS home-based parent program ($n = 314$) and the other half were assigned to a 'No PALS' control condition ($n = 309$). This 2×2 research design resulted in 623 parent-child dyads nested within 77 classrooms with four conditions: (1) TEEM/PALS; (2) TEEM/No PALS; (3) No TEEM/PALS, (4) No TEEM/No PALS; see Fig. A1 in Supplementary materials. The sample after attrition is shown in Table 1.

2.2. Participant characteristics

Children's average age was 4.37 years (range = 3.18–5.99), and age did not differ across conditions. Children were balanced according to gender (50% female). Most children were Latino/Hispanic (66%) or African American (28%) and almost half spoke English in the home (48%). However, 35% of the sample spoke Spanish in the home, and 17% spoke both English and Spanish. Parents reported that 96% of children qualified for free lunch. The primary caregiver's highest reported level of education varied substantially, with 27% having achieved a high school diploma or GED and 21% having completed some college coursework (see Table 1).

Teachers' background characteristics are shown in Table 1. Most teachers were female (93%) and of Latino/Hispanic (45%) or African American (45%) ethnicities. Teachers reported an average of 10 years of teaching experience, although there was considerable variability ($SD = 8.69$; range 0–39 years). Most teachers held a bachelor's degree or higher level of education. Most teachers spoke English in the classroom (62%) and some spoke Spanish to varying degrees (29% half English/half Spanish; 8% mostly Spanish; 2% Spanish).

2.3. Teacher professional development

Teachers randomly assigned to the TEEM condition participated in several professional development activities including: (a) two-day initial training, (b) in-class coaching, (c) coursework, (d) progress monitoring, and (e) provision of instructional resources. The initial two-day training introduced teachers to components of the TEEM framework and to all intervention topics including: classroom organization, setting up a schedule, language development, phonological awareness, reading books to support listening

comprehension and vocabulary, letter knowledge and mathematics instruction. Coaches provided individualized in-class coaching on the full range of topics, even if that topic had not yet been addressed in the coursework.

The in-class coaching occurred in fourteen 2-h sessions approximately biweekly (4 h per month). Trained coaches used a variety of strategies (e.g., side-by-side coaching, modeling lessons) to support teachers in implementing evidence-based instructional techniques. The coach and teacher also met for 15–30 min for a reflective follow-up conversation with feedback on strengths and areas to improve, including the use of progress monitoring data to inform instruction. The coursework included between 16–20 weekly afternoon sessions for 2 h. each through a platform called eCIRCLE. Variability in number of weeks occurred because some cohorts and sites began the project later than expected and had to complete their coursework in a condensed time frame. Nonetheless, the same content was covered for all teachers by condensing content into fewer face-to-face sessions as needed. During each session, the coach used web-based courses to facilitate group discussion and role play of evidence-based instructional techniques. Teachers were provided with a Lenovo netbook computer to ensure access to the web-based courses, complete "homework" activities, and administer progress monitoring measures. The topics and timing of sessions are provided in Table A3 in Supplementary material.

During an early coursework session, teachers were trained to administer progress monitoring measures from the CIRCLE Screening Assessment and Progress Monitoring System (Landry, Swank, Assel, & King, 2009) three times during the academic year (beginning-, middle- and end-of-year). The progress monitoring measures consisted of the following subtests: (a) 60-s letter naming fluency, (b) 60-s vocabulary fluency, (c) untimed, 43-item phonological awareness screen comprised of seven tasks, (d) untimed, 27-item mathematics tasks, (e) 10 observational ratings of book and print awareness, (f) 11 observational ratings of early writing skills, and (g) seven observational ratings of social skills. Teachers received incentive pay for their participation and completion of study related measures and questionnaires.

Instructional resources were provided to teachers in the TEEM condition, and other materials were provided across the TEEM and No TEEM conditions. First, if needed, a core curriculum from the state approved list was purchased for all classrooms according to the choice of the school/district. Most schools already had a core curriculum (e.g., Scholastic Big Day for PreK, DLM Let's Begin with the Letter People, Frog Street Press) so no additional core curricula were provided. Second, supplemental kits were provided to all TEEM teachers including materials to support classroom management (HATCH "Positive Beginnings" kit) and school readiness instructional materials (Lakeshore "Ready to Read Toolkit").

2.4. Parent intervention

2.4.1. PALS condition

Parents randomly assigned to the PALS treatment condition participated in a 19-session home-based responsive parenting intervention: the Play and Learning Strategies (PALS) program. Each family in this condition was assigned a Family Coach who implemented the PALS curriculum with the primary caregiver (most often the mother) through one-on-one sessions across the school year (approximately weekly, with breaks built in for holidays). Sessions took place in the home unless the family requested a different location (e.g., grandparent's home, public library meeting room). The PALS curriculum is guided by a detailed manual and a set of videotapes that help the parent learn to use responsive behaviors (e.g., attending to and understanding young children's communicative signals, responding promptly and contingently, following the child's lead in play and maintaining rather than redirecting

the child's focus of attention, using scaffolding strategies to support children's language development, and using positive behavior management strategies) during play and learning activities (e.g., shared book reading) and daily routines (e.g. mealtime, dressing).

Although PALS focuses on one designated parent throughout the program, the child's other parent or caregiver is welcomed to attend sessions. Two sessions (one midway, one near the end of the program) focus specifically on reviewing and sharing of PALS skills with a second caregiver (e.g., father, grandparent) whom the participating parent invites to join the session. These sessions allow the coach to informally assess the parent's ability to articulate and demonstrate what she has been learning, as well as familiarizing the second caregiver with the target strategies. PALS has been validated in several previous random-assignment efficacy studies which have shown its significant positive effects on maternal responsiveness behaviors as well as children's language skills, cooperation, and social engagement (e.g., Landry et al., 2008). In this study, PALS sessions ran on a timeline parallel to the TEEM teacher intervention (see online Supplement Table A3). Weekly sessions, lasting approximately 1–1.5 h each, alternated between didactic sessions in which new topics were introduced and discussed with video exemplars of parents demonstrating the behaviors, and coached practice sessions in which the parent was guided by the coach to implement new and previously learned strategies during play time or a daily routine activity with her child. Parents were videotaped during this practice and asked to self-reflect with guidance from their coach. The PALS curriculum is available in Spanish and English versions and was delivered in the parent's self-designated primary language by a coach fluent in that language. PALS was implemented in English for 54.5% of families and in Spanish for 45.4%.

2.4.2. No PALS condition

Parents randomly assigned to the No PALS control condition, along with parents in the PALS condition, were sent a series of 10 written handouts (1–2 pages in length), spaced throughout the school year, each covering a general preschool child-rearing topic such as developmental milestones for 4-year-olds, simple games to promote children's learning of letter and numbers, guiding children's TV viewing, helping children cope with fears, and responding to sibling conflicts. Handouts were sent home from school with the child and were provided in English or Spanish according to parent preference.

2.5. Characteristics of classroom coaches

Two senior staff with experience conducting the TEEM program implemented all teacher professional development and coaching components for the TEEM condition. Both coaches completed a Coaching Institute in September of each academic year and followed manualized procedures for delivering the intervention. One coach was assigned to all the Houston teachers across all cohorts and the other coach was assigned to all the Austin teachers across cohorts. Both coaches were Caucasian, monolingual English speakers who held a bachelor's degree and had over 25 years of early childhood education experience. TEEM project managers supervised coaches monthly by reviewing implementation reports.

2.6. Characteristics of parent coaches

Coaches who conducted the PALS intervention with families ($n=21$: 18 in Houston and 3 in Austin) were all female and were ethnically diverse (13 Hispanic/Latina, 4 African American, and 4 Caucasian). Seventeen of the coaches were bilingual, and all coaches had attained a bachelor's degree. Prior to working with families,

coaches completed comprehensive training with a senior research staff that included a standardized certification process.

2.7. Fidelity of implementation

Fidelity was assessed for teachers, parents, and coaches; these data are detailed in online Supplementary Table A4. Across approximately four observations, classroom coaches' use of 3-point rating scales of adherence indicated moderate TEEM implementation by teachers in using: provided materials ($M=2.07$), suggested cognitive instruction approaches ($M=2.02$), and a responsive style ($M=2.15$). To assess dosage and duration of professional development activities, we recorded teacher attendance at coursework and coaching sessions. All of the teachers attended the initial 2-day training. On average, teachers received 79.41% of the coursework sessions and 91.93% of the coaching sessions.

For parents, a series of 3-point fidelity ratings examined the extent to which parents were engaged in the activities and content of PALS sessions. Across the 19 PALS sessions parents typically demonstrated high engagement ($M=2.86$) and moderate-high mastery of skills taught ($M=2.54$). In terms of dosage, on average, parents participated in 77.89% of the sessions. Coaches were supervised regularly by senior research staff to ensure fidelity, including weekly group supervision meetings and monthly home visits.

2.8. Assessment procedures

To examine changes in teacher behaviors, a classroom observation was conducted at pretest (August–September), midyear (December–January) and posttest (April–May). To assess differences in parenting practices, parents were videotaped at pretest and posttest playing with and reading to their child, using standardized protocols. Specifically, during a home visit, examiners provided parents with a standard set of toys (i.e., blocks, dishes, truck, toy animals, play dough) and asked them to play with their child as they normally would for 10 min. Then, examiners provided parents with a standard set of six books (e.g., *Cat in the Hat*; *The Very Hungry Caterpillar*) in either English or Spanish and asked parents to share the books with their child as they normally would for 10 min.

At pretest and posttest, children were assessed in a quiet location within their schools over four sessions, and teachers and parents completed rating scale measures of children's social, behavioral and self-regulatory functioning. Assessment procedures were coordinated by an author of the study, who conducted initial trainings and annual refresher trainings for the examiners. Lab certification was required for all assessment protocols prior to conducting testing in the field, and all examiners were blind to condition.

Children were assessed with measures of language, early literacy, and self-regulation skills. The self-regulation tasks were administered in either English or Spanish. In addition to the standard battery, some children ($n=222$) also completed Spanish language and literacy measures, including the Expressive One-Word Picture Vocabulary Test – Spanish-Bilingual Edition (EOWPVT-SBE; Brownell, 2001) and the Spanish version of the Preschool Language Scale – Fourth Edition (PLS-4; Zimmerman, Steiner, & Pond, 2002). Criteria for completing these Spanish assessments included: (a) receipt of the parenting intervention in Spanish, or (b) indication from parents in the control condition that their dominant home language was Spanish. Administration of Spanish assessments was nearly equal across conditions ($ns=54-57$).

2.9. Measures

2.9.1. Teaching quality

The Teacher Behavior Rating Scale (TBRS; Landry, Crawford, Gunnewig, & Swank, 2001) was used to evaluate teacher responsiveness and instruction. Coding for the TBRS was conducted live, by field certified TBRS coders. This certification included demonstrating mastery of codes, achieving reliability coding videotapes, and then conducting side-by-side live coding with master coders ($\geq 80\%$ agreement with the master coder prior to independent coding). Finally, 15% of live observations were double coded for reliability checks with annual retraining to minimize drift and errors in coding. Observers made ratings across two hours in the classroom during which they observed a range of activities such as circle time, centers, transitions, and small-group activities. Items, rated for both quality and quantity, are included in 11 TBRS subscales: Sensitivity (12 items), Classroom Community (5 items), Oral Language (7 items), Book Reading Behaviors (8 items), Phonological Awareness (2 items), Print and Letter Knowledge (7 items), Written Expression (3 items), and Math (5 items), Team Teaching (5 items), Centers (7 items), Lesson Plans (8 items; see online Supplement Table A5 for detailed descriptions). Quantity ratings are made on a 4-point scale, with 1 = *never* and 4 = *often*. Quality ratings are made on a 4-point scale, with 1 = *low* and 4 = *high*. For each subscale, quality and quantity scores were averaged to compute the subscale score used in analyses. The subscale scores were averaged to compute a TBRS total score (see Table 2 descriptives). The TBRS has previously shown high reliability and validity (e.g., interrater reliability ranged from 0.71 to 1.00; Landry, Crawford et al., 2001; Landry, Smith et al., 2001). The Cronbach alphas of the subscale scores in the current study were 0.85–0.87 for the three waves of data collection.

2.9.2. Parenting quality

Ratings of parenting quality and frequency counts of specific parental behaviors were made using the videotaped parent-child free play and book reading sessions. Coding of the videotaped parent-child free play and book reading sessions was supervised by a master coder under the direction of the first author. Bilingual coders, who were blind to intervention status, first spent three weeks in training to achieve reliability ($\geq 80\%$ agreement with the master coder). Monthly meetings were held to guard against observer drift. During these meetings, videotapes were coded as a group, and interrater agreements were checked to ensure that they met the criterion of $\geq 80\%$ per variable. A second rater coded approximately 15% of the videotapes, and interrater reliability was examined by calculating intraclass correlations (ICCs), which are reported below for each coded measure.

2.9.2.1. Ratings. Ratings of parenting quality were made on a 5-point scale with higher values indicating better parenting quality (e.g., 1 = *almost never*; 5 = *almost always*). Full descriptions of the rating scales are provided in online Supplementary Tables A6 and A7. These scales have been used extensively in previous research and found to have high reliability and validity (e.g., Landry et al., 2008). All rating scales in the present data were factor analyzed to identify overlap and create composite variables (Cronbach's $\alpha = 0.73$ – 0.75).

2.9.2.2. Frequencies. For the book reading session, we coded the frequencies of specific parental behaviors, including the use of prompts to encourage verbal responses, praise, verbal scaffolding, tracking print with child, and time spent just reading the text (see Table A6 in Supplementary materials). For the free play session (see Table A7 in Supplementary materials), coded parental behaviors included the use of prompts for child verbal responses, provision of verbal labels, praise, provision of verbal scaffolding, maintain-

ing attention, and redirecting child attention away from a toy or activity of interest (ICC = 0.64–0.87).

2.9.3. Standardized child literacy and language measures

The Test of Preschool Early Literacy (TOPEL; Lonigan, Wagner, Torgesen, & Rashotte, 2007) consists of three subtests: (1) Print Knowledge – 36 items; e.g., letter recognition, book and print concepts; (2) Phonological Awareness – 27 items; e.g., elision and blending tasks; (3) Definitional Vocabulary – 35 items; e.g., name and answer questions about pictures. Internal consistency was high: Print Knowledge ($\alpha = 0.97$ pretest, 0.96 posttest), Phonological Awareness ($\alpha = 0.88$ pretest, 0.90 posttest), and Definitional Vocabulary ($\alpha = 0.98$ pretest, posttest).

The Expressive One-Word Picture Vocabulary Test (EOWPVT; Brownell, 2001), is a norm-referenced test that measures expressive vocabulary. Children are presented with a series of illustrations depicting an object, action, or concept and asked to name each illustration. Internal consistency (Cronbach's α) was 0.97 at both pretest and posttest.

The Preschool Language Scales – Fourth Edition (PLS-4; Zimmerman et al., 2002), is a global oral language measure consisting of the Auditory Comprehension scale assessing receptive language and the Expressive Communication scale assessing the child's ability to communicate with others. Cronbach's α was 0.94 at pretest and 0.93 at posttest for Auditory Comprehension; 0.96 at pretest and posttest for Expressive Communication.

Due to high correlations among some language measures from the children's cognitive battery ($r_s = 0.62$ – 0.86), we formed a composite “language skills” factor consisting of the EOWPVT, PLS-Auditory Comprehension, PLS-Expressive Communication, and TOPEL-Definitional Vocabulary measures. Internal consistency (Cronbach's α) was high at both pretest (0.93) and posttest (0.92).

2.9.4. Child social-behavioral functioning

Parents and teachers completed the Children's Behavior Questionnaire (CBQ; Rothbart, Ahadi, Hershey, & Fisher, 2001) Attention Focusing (14 items; e.g., *when picking up toys, usually keeps at the task until it is done*) and Inhibitory Control (13 items; e.g., *can wait before entering into new activities if asked*) scales, which fall into the effortful control dimension (Rothbart et al., 2001). Items are rated on a scale ranging from 1 (*extremely untrue of your child*) to 7 (*extremely true of your child*). Raw scores on these two scales were combined to form an effortful control composite (after reverse scoring items if appropriate). The CBQ scales have demonstrated internal consistency ranging from 0.64 to 0.92 (average $\alpha = 0.77$) and interrater reliability estimates ranging from 0.67 to 0.92 (average $\alpha = 0.75$; Rothbart et al., 2001). In the current study, for the parent-rated effortful control composite, internal consistency (Cronbach's α) was 0.81 at pretest and 0.83 at posttest. For the teacher-rated effortful control composite, Cronbach's α was 0.90 at pretest and 0.92 at posttest.

Parents and teachers completed the Social Competence and Behavior Evaluation (SCBE-30; LaFreniere & Dumas, 1996) that contains three subscales with 10 items each: Social Competence (e.g., *cooperates with other children*), Anger/Aggression (e.g., *easily frustrated*), and Anxiety/Withdrawal (e.g., *avoids new situations*). Items are rated on a scale ranging from 1 (*never*) to 6 (*always*). Scores on the three scales were averaged to create total scores. In this sample, for the parent-rated variables, internal consistency (Cronbach's α) was adequate for Social Competence (0.80 at pretest and 0.81 at posttest), Anger/Aggression (0.81 at pretest and 0.79 at posttest), and Anxiety/Withdrawal (0.74 for pretest and posttest). For the teacher-rated variables, Cronbach's α was high for Social Competence (0.88 at both pretest and posttest), Anger/Aggression (0.93

Table 2

Descriptive statistics and results of linear growth models examining effects of TEEM on change in TBRS scores over time.

	No TEEM – <i>M</i> (<i>SD</i>)			TEEM – <i>M</i> (<i>SD</i>)			Pretest	Mid-year	Posttest	
	Pretest	Mid-year	Posttest	Pretest	Mid-year	Posttest	<i>b</i>	<i>b</i>	<i>d</i>	<i>d</i>
Book Reading Behaviors	2.0 (0.4)	1.9 (0.4)	2.1 (0.5)	2.0 (0.5)	2.4 (0.6)	2.3 (0.5)	0.12	0.42***	0.90	0.19
Learning Centers	2.1 (0.6)	2.2 (0.6)	2.6 (0.8)	2.2 (0.6)	2.5 (0.7)	3.0 (0.6)	0.11	0.36*	0.57	0.43**
Classroom Community	2.4 (0.6)	2.5 (0.5)	2.7 (0.5)	2.4 (0.6)	2.8 (0.5)	2.9 (0.5)	0.04	0.27*	0.45	0.23†
Teacher Sensitivity	1.9 (0.3)	1.9 (0.4)	2.0 (0.4)	2.0 (0.4)	2.2 (0.4)	2.2 (0.5)	0.09	0.24*	0.68	0.16
Lesson Plans	2.0 (0.9)	2.6 (0.7)	2.6 (0.9)	2.2 (0.7)	2.9 (0.7)	3.0 (0.8)	0.23	0.38*	0.47	0.38*
Math Concepts	1.9 (0.6)	1.8 (0.6)	1.8 (0.6)	1.8 (0.6)	2.2 (0.7)	2.1 (0.7)	−0.05	0.41**	0.67	0.20
Oral Language	2.1 (0.4)	2.1 (0.5)	2.3 (0.5)	2.2 (0.4)	2.4 (0.5)	2.5 (0.6)	0.10	0.29*	0.66	0.21†
Phonological Awareness	2.6 (0.1)	2.6 (0.1)	2.6 (0.1)	2.6 (0.1)	2.6 (0.2)	2.7 (0.2)	–	–	–	–
Print and Letter Knowledge	2.0 (0.5)	2.2 (0.5)	2.3 (0.5)	1.9 (0.4)	2.4 (0.6)	2.6 (0.6)	−0.09	0.23†	0.48	0.33**
Written Expression	1.7 (0.5)	1.8 (0.6)	1.9 (0.6)	1.8 (0.6)	2.0 (0.5)	2.1 (0.5)	0.12	0.20	0.38	0.20
Total	2.0 (0.3)	2.1 (0.3)	2.3 (0.3)	2.1 (0.3)	2.4 (0.3)	2.5 (0.3)	0.07	0.31***	0.93	0.24**

Note. TEEM, The Early Education Model; TBRS, Teacher Behavior Rating Scale. The model for Phonological Awareness would not converge due to lack of variability.

† $p < 0.10$.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

at pretest and 0.90 at posttest), and Anxiety/Withdrawal (0.83 at pretest and 0.80 at posttest).

2.9.5. Child school liking and avoidance

Teachers completed the *School Liking and Avoidance Questionnaire* (SLAQ; Ladd & Dinella, 2009), which asked them to rate children on 13 items using a scale ranging from 1 (*almost never*) to 5 (*almost always*). The SLAQ consists of two subscales: school avoidance (6 items; e.g., *complains about school*) and school liking (7 items; e.g., *enjoys most classroom activities*). Internal consistency was $\alpha = 0.82$ – 0.85 at pretest and $\alpha = 0.78$ – 0.83 at posttest. More positive scores, reflecting greater comfort with classroom experiences, were expected to align with increased support from teachers and parents for school readiness skills.

2.9.6. Child self-regulation

Children completed two tasks measuring “hot” self-regulation. In the gift delay-wrap task (e.g., Kochanska, Murray, & Harlan, 2000), children were told that they would receive a present but that they could not turn and peek while the present was being noisily wrapped by the examiner for 60 s. Strategy scores (1 = leaves seat to peek, 2 = turns body in seat to peek, 3 = peeks over shoulder, 4 = does not peek) given for every 15 s of the task were averaged to create a total strategy score for the task. Scores for strategy and latency to peek (in seconds), which were strongly correlated ($r = 0.76$ at pretest; $r = 0.78$ at posttest), were standardized and averaged to create a total score ($ICC = 0.94$).

In the gift delay-bow task (Kochanska et al., 2000), the wrapped gift was placed in front of the child and the child was told not to touch the gift until the examiner returned with a bow. The delay lasted two minutes. Strategy scores (1 = removes toy from box or touches toy in box, 2 = lifts the lid of the box ≥ 3 in. or takes lid off, 3 = lifts the lid of the box ≤ 2 in., 4 = touches the box but does not open it, 5 = does not touch the box) given for every 15 s of the task were averaged to create a total strategy score for the task. Scores for strategy and latency to touch (in seconds), which were strongly correlated ($r = 0.62$ at pretest; $r = 0.63$ at posttest), were standardized and averaged to create a gift delay-bow total score (interrater reliability: $ICC = 0.99$).

Children completed one task measuring “cool” self-regulation. In the bear/dragon task (Carlson, 2005; Garon et al., 2008), children were told to follow the “nice” bear puppet’s commands (e.g., touch your nose) but not the “mean” dragon puppet’s commands. After practicing, there were 12 test trials (two sets of 6 trials, with a reminder of the rules in the middle) with the bear and dragon commands alternating in pseudo-random order. Trial scores rang-

ing from 0 to 2 (0 = no movement, 1 = partial movement, 2 = full movement) were summed for the bear and dragon separately. Performance on dragon trials served as the outcome measure; a pass/fail score was used in analyses based on recommendations from previous studies (5/6 correct dragon trials constituted passing the task; Carlson, 2005). Interrater reliability was high ($ICC = 0.99$), and Cronbach’s α for the dragon trials was 0.97 (pretest) 0.96 (posttest).

2.10. Attrition and missing data

We examined whether attrition differed substantially between groups, as well as whether participants who attrited differed on the outcome variables at pretest.

2.10.1. Teacher

Of the original 79 teachers recruited, 70 (89%) remained in the study through posttest. Of the nine who dropped out, two were never replaced, six were replaced in time to collect data from them at pretest, and one was replaced too late in the year for a pretest observation. Follow-up interviews revealed that teachers left the study for reasons unrelated to the intervention, including classroom dissolution, transfer to a different classroom, termination, and leaving the school. Analyses comparing teachers who attrited to those who stayed revealed no significant differences on the TBRS at pretest, suggesting that missingness at the teacher level was ignorable.

2.10.2. Parent-child

Of the 623 families and children recruited for the study, 434 completed the study, for a 30% rate of attrition. Rates of attrition were similar across conditions, and analyses of the child and parent outcomes (e.g., cognitive, social, and self-regulation, and parent responsiveness variables) revealed only one significant difference at pretest between children who left the study and children who remained in the study: school liking (4.43 – remained; 4.28 – left) and one parent variable (redirection during parent-child toy play; 1.28 – remained; 0.73 – left). Parents who left the study were comparable on level of education and ethnicity compared to those that remained. As parents that remained showed more redirection (a negative behavior) at pretest than those that left, and the difference in school liking was relatively small, there was little evidence that the patterns of missing data were non-ignorable and likely to bias the results (Little & Rubin, 2014). Attrition rates for all outcomes are in Tables A1–A2 of the online Supplemental materials.

3. Results

3.1. Pre-analysis data inspection

Pre-analysis data inspection included examination of patterns of missing data, potential outliers, and departures from normality. Mild departures from normality were found for several variables. Non-normality of variables was handled using different model assumptions (e.g., Poisson, negative binomial, logistic). In all analyses, maximum likelihood estimation was employed, which yields unbiased estimates in the presence of missing data when the pattern of missingness is ignorable, which is likely given the lack of any systematic differences at pretest between those who completed the study and those who attrited (Little & Rubin, 2014). Growth analyses of the teacher-level TBRS outcomes used all available information (FIML), and ANCOVA analyses of parent and child outcomes used the data from those participants who were assessed at both time points.

3.2. General analytic strategy

To address our three research questions we first examined the effect of the TEEM program on teacher behaviors. Next, we considered the effect of the PALS program on parent behaviors. Finally, we examined the impact of each intervention on children's outcomes, including their separate and combined effects.

Teachers were assessed at three time points, so a growth modeling perspective was used to examine TEEM effects on change in TBRS scores over time (Raudenbush & Bryk, 2002). Parents and children were assessed at pretest and posttest, so an ANCOVA model was used to assess posttest differences controlling for levels at pretest. Child-level data were nested within classrooms, so multilevel modeling was employed when appropriate to control for the non-independence of observations. For the child-level analyses, the interaction of TEEM and PALS was entered in the model with the main effects of TEEM and PALS first, and if not significant, the interaction was removed and the main effects were tested simultaneously in a subsequent model. Cohen's *d* was calculated as a measure of effect size.

3.3. Teacher outcomes

Descriptive statistics for the three time points (pretest, midyear, posttest) of the TBRS can be found in Table 2. Examination of these means over time revealed differences in the growth trajectories between the first and second time points and the second and third time points. Therefore, a piecewise linear growth strategy was used to analyze TEEM effects on teaching quality (TBRS scores) across the three time points. Piecewise linear regression is similar to growth curve modeling, except that separate slopes are modeled between the first two time points ("phase 1") and the last two time points ("phase 2"). This model is most appropriate when there is a sharp break in the trajectories between phases of the intervention that cannot be captured by a linear or curvilinear growth model.

We first tested for pretest differences between the TEEM and control teachers, and found no differences (see Table 2). Results indicated that the group by time interaction was significant for change from pretest to midyear for the TBRS total score. Teaching quality was higher for TEEM teachers compared to control teachers at the midyear time point of the study (*d* = 0.93). Also, most of the subscales were significantly higher for the TEEM as compared to the control teachers from the pretest to the midyear measure (see Table 2). The group by time interaction for change from midyear to posttest also was significant for the total score. TEEM teachers

Table 3
Effects of PALS on parenting behaviors during book reading and free play sessions.

Frequencies	Book reading		Free play	
	<i>b</i>	<i>d</i>	<i>b</i>	<i>d</i>
Prompts	0.60*	0.24	3.89***	0.44
Praise	0.48*	0.25	1.32***	0.53
Verbal scaffolding	0.32**	0.32	0.14*	0.18
Text reading duration	−23.89***	−0.59	–	–
Tracking print	1.45***	0.38	–	–
Provision of labels	–	–	3.38***	0.32
Maintain attention	–	–	−0.92	−0.19
Redirect attention	–	–	−0.72***	−0.41
Rating Scales				
Responsive language and behavior support	1.33***	0.44	1.23***	0.48
Language building strategies	0.41***	0.55	–	–
Negativity	2.53**	−0.56	0.45**	−0.49

Note. For book reading, 'responsive language and behavior support' is a composite score composed of the following scales: enthusiasm and engagement, responsiveness/flexibility, language comprehension techniques. For free play, 'responsive language and behavior support' is a factor score composed of the following scales: responsiveness/flexibility, positive affect, warmth, and verbal content. Parent results are from analyses using complete data.

* *p* < 0.05.

** *p* < 0.01.

*** *p* < 0.001.

had higher scores compared to control teachers at the posttest time point (*d* = 0.71).

3.4. Parent outcomes

Analyses of the parent outcomes were conducted using multi-level ANCOVA (see Table 3 for these results). Although we did not expect any influence of the TEEM intervention on parent outcomes, in order to examine treatment specificity, we used the same procedures for the parent outcomes that we did for all others, including the tests of the interaction and main effect of TEEM. For the parent outcomes, as expected, there were no significant effects for TEEM or the interaction.

3.4.1. Behaviors during shared book reading

For ratings of parenting quality during shared book reading, there were significant main effects of PALS. Specifically, parents who received PALS were rated higher on the Responsive Language and Behavior Support composite (*d* = 0.44) and on Language Building Strategies (*d* = 0.55), and lower on the intrusiveness and negative tone scale (*d* = −0.56) compared to parents who did not receive PALS. For coded frequencies of specific parenting behaviors during shared book reading, there were numerous significant main effects of PALS. Parents who received PALS showed greater gains in using prompts to encourage verbal responses (*d* = 0.24), praising their child (*d* = 0.25), verbal scaffolding (*d* = 0.32), and tracking print with their child (*d* = 0.38) as well as spending less time just reading the text (*d* = −0.59) compared to parents who did not receive PALS (see Table 3).

3.4.2. Behaviors during free play

For ratings of parenting quality during free play, there were significant main effects of the PALS intervention. Parents who received PALS displayed greater gains on the Responsive Language and Behavior Support Composite (*d* = 0.48) and had greater decreases in negativity (*d* = −0.49) compared to parents who did not receive PALS. Similarly, for observed frequencies of parenting practices during free play, there were also significant findings in favor of parents who received the PALS program. PALS parents exhibited greater gains in the use of prompts for children to give a verbal response (*d* = 0.44), provision of verbal labels (*d* = 0.32), praise

Table 4
Effects of TEEM and PALS on children's behavior during book reading and free play sessions.

Book reading scales	Predictor	<i>b</i>	<i>d</i>
Book reading engagement	TEEM × PALS	0.63 [*]	0.54
	TEEM vs. no TEEM	0.21	0.18
	PALS vs. no PALS	0.32 [*]	0.27
Shared enjoyment	TEEM × PALS	0.40	0.32
	TEEM vs. no TEEM	0.26 [*]	0.21
	PALS vs. no PALS	0.35 ^{**}	0.28
Language use	TEEM × PALS	0.30	0.24
	TEEM vs. no TEEM	0.25	0.20
	PALS vs. no PALS	0.35 [*]	0.28
Enthusiasm/initiative	TEEM × PALS	1.32	0.42
	TEEM vs. no TEEM	0.72 [*]	0.23
	PALS vs. no PALS	1.01 ^{**}	0.32
Free play scales	Predictor	<i>b</i>	<i>d</i>
Cooperation	TEEM × PALS	0.75	0.18
	TEEM vs. no TEEM	1.12	0.07
	PALS vs. no PALS	0.83	0.12
Social engagement	TEEM × PALS	0.96	0.02
	TEEM vs. no TEEM	1.06	0.04
	PALS vs. no PALS	0.88	0.08

Note. Child results are from analyses using complete data.

**p* < 0.05.

***p* < 0.01.

****p* < 0.001.

(*d* = 0.53), provision of verbal scaffolding (*d* = 0.18) and greater decreases in redirecting children's attention away from a toy or activity in which they showed interest (*d* = −0.41) (see Table 3).

3.5. Child outcomes

For the child outcomes, multilevel ANCOVAs were conducted to examine the main effects of TEEM and PALS as well as their interaction on posttest scores, controlling for pretest scores, child home language, and child age (see results in Tables 4 and 5; descriptive statistics in online Supplement Table A8).

3.5.1. Behaviors during book reading and free play

Results for children's observed social and linguistic behaviors are presented in Table 4. During book reading, there was a PALS vs. No PALS effect (*d* = 0.27) and a TEEM by PALS interaction for book engagement (*d* = 0.54). PALS children showed greater engagement than children in the No PALS condition, and children in the PALS condition who were also in a TEEM classroom showed even greater engagement than those in the other three conditions. There were also significant TEEM vs. No TEEM findings for shared enjoyment (*d* = 0.21) and initiative (*d* = 0.23) and findings for PALS vs. No PALS for language use (*d* = 0.28), shared enjoyment (*d* = 0.28) and initiative (*d* = 0.32). There were no significant group differences for children's cooperation or social engagement during free play.

3.5.2. Literacy and language

As shown in Table 5, PALS children scored significantly higher on the TOPEL Print Knowledge subtest than children whose parents did not receive PALS (*d* = 0.14). This result demonstrated that parents receiving PALS, whether their children were in a TEEM or control classroom, were providing stimulation that promoted the development of print and letter knowledge. There was also a TEEM by PALS interaction (*d* = −0.33), demonstrating that children who were in TEEM classrooms and whose parents were not in the PALS program made greater gains on the TOPEL Phonological Awareness subtest than children in the other groups, who did not differ

Table 5
Effects of TEEM and PALS on children's outcomes.

School Readiness Outcomes	Predictor	<i>b</i>	<i>d</i>
Cognitive skills			
Print Knowledge	TEEM × PALS	0.38	0.03
	TEEM vs. no TEEM	1.08	0.10
	PALS vs. no PALS	1.53 [*]	0.14
Phonological Awareness	TEEM × PALS	−1.69 [*]	−0.33
	TEEM vs. no TEEM	0.49	0.10
	PALS vs. no PALS	−0.18	−0.04
Language skills	TEEM × PALS	−0.14	−0.16
	TEEM vs. no TEEM	0.04	0.05
	PALS vs. no PALS	−0.16	−0.18
Social and behavioral skills			
Effortful Control – Parent	TEEM × PALS	−0.01	−0.02
	TEEM vs. no TEEM	−0.01	−0.02
	PALS vs. no PALS	0.10 [*]	0.15
Effortful Control – Teacher	TEEM × PALS	−0.05	0.05
	TEEM vs. no TEEM	−0.03	−0.03
	PALS vs. no PALS	0.07	0.07
Social Competence – Parent	TEEM × PALS	0.096	0.18
	TEEM vs. no TEEM	0.026	0.05
	PALS vs. no PALS	0.070	0.14
School Liking	TEEM × PALS	−0.12	0.20
	TEEM vs. no TEEM	0.14 ⁺	0.23
	PALS vs. no PALS	0.07	0.12
School Avoidance	TEEM × PALS	0.07	0.12
	TEEM vs. no TEEM	−0.14 [*]	−0.24
	PALS vs. no PALS	−0.04	−0.06
Self-regulation skills			
Gift delay-wrap task	TEEM × PALS	0.001	0.01
	TEEM vs. no TEEM	−0.031	−0.05
	PALS vs. no PALS	0.16 [*]	0.19
Gift delay-bow task	TEEM × PALS	0.143	0.18
	TEEM vs. no TEEM	−0.024	−0.03
	PALS vs. no PALS	0.047	0.06
Bear/dragon task	TEEM × PALS	−0.103	−0.21
	TEEM vs. no TEEM	0.007	0.01
	PALS vs. no PALS	0.05	0.09

Note. The 'language skills' factor is composed of the Expressive One-Word Picture Vocabulary Test, Preschool Language Scale-4 Auditory Comprehension, Preschool Language Scale-4 Expressive Communication, and Test of Preschool Early Literacy Definitional Vocabulary. Child results are from analyses using complete data.

**p* < 0.05.

***p* < 0.01.

****p* < 0.001.

from each other. There was no difference between groups on the language composite.

3.5.3. Social and behavioral skills

There was a significant difference between the PALS and NoPALS groups for parent-reported effortful control (*p* = 0.03, *d* = 0.15). Parents in the PALS condition reported greater effortful control in their children than parents in the NoPALS condition. There were no significant differences across the groups for parent or teacher report on the SCBE-30.

There was a significant main effect of TEEM indicating that children in the TEEM classrooms showed greater decreases in school avoidance compared to children in control classrooms (*p* = 0.03, *d* = 0.24). There was also a similar, though non-significant, effect (*d* = 0.23; *p* = 0.06) for children in the TEEM classrooms to show greater gains in school liking compared to children in the control classrooms.

3.5.4. Self-regulation

For the gift delay-wrap task, there was a significant main effect of PALS, *p* = 0.04, *d* = 0.19. Children whose parents received PALS were better able to withhold the urge to peek while the gift was being wrapped than children whose parents did not receive PALS. For the gift delay-bow task, there was a small but non-significant

effect in the expected direction for a TEEM by PALS interaction ($p=0.16$, $d=0.18$). Regarding cool self-regulation, there were no group differences on the bear/dragon task.

4. Discussion

This study examined whether training both teachers and parents to use an emotionally- and cognitively-responsive style would result in greater school readiness skills for children enrolled in Head Start programs than either intervention alone or a control group. Results showed significant increases in targeted adult behaviors across the school and home. Some intervention effects were observed for children's school readiness skills, but we consider below possible reasons why group differences did not emerge in all areas.

4.1. Effect of the interventions on parent and teacher behaviors

Based on our theory of change, it was important to improve parents' and teachers' use of responsive behaviors, as it was through these changes that we hypothesized children would increase in their school readiness skills. Similarities in many characteristics of the two interventions may help explain observed increases for both parents and teachers. Both interventions incorporated strategies supported by research on adult learning (e.g., Bransford, Brown, & Cocking, 2000), including the use of coaching. Coaches acknowledged parents' and teachers' beliefs concerning their role in supporting young children's school readiness and provided active engagement in understanding the theory behind and the application of the new responsive practices introduced. Both PALS and TEEM also used many video exemplars of responsive behaviors in all intervention sessions with believable models in authentic settings. Informal feedback about these videos suggested that seeing and talking about video models with their coach was a key to facilitating increases in parents' and teachers' use of new behaviors, as they could see the positive impact of the behaviors on children's responses. Research also supports the need to provide opportunities for training and practice over time rather than during one workshop (Ball & Cohen, 1999); both TEEM and PALS incorporated ongoing supports.

In addition to ongoing use of training, coaching, and video supports, other characteristics of the interventions that were potentially important for understanding changes in the parent and teachers' target behaviors were: (a) coaches' use of measurement tools to guide goal setting, and (b) parents and teachers engaging in varying degrees of self-reflection. In TEEM, coaches used a fidelity checklist to guide their coaching, while in PALS, coaches filled out an engagement checklist after each session to help them evaluate parents' understanding. In both cases, more effective use of this data or other measurement tools might have resulted in greater changes in behavior. For example, our coaches may not have always followed through to complete the final step of setting goals for improvement based on teachers' and parents' areas of need, a common problem reported by others (Neuman & Wright, 2010). The wide range of the fidelity scores for TEEM and PALS evidence large variability in achieving good implementation of the two interventions (see online Supplemental Table A4). Although self-reflection was part of both interventions, it occurred differently in the two interventions. In TEEM, self-reflection only occurred through a brief follow-up conversation between the teacher and their coach after the coaching session. In PALS, self-reflection was a more central component, as parents were videotaped practicing each new behavior with their child at each coaching session and then guided by their coach to view themselves and self-critique their behavior. Most parents were able to talk about what they thought they did that helped

their child and what was not helpful. For those who had difficulty doing this, the coach cued them to notice certain behaviors. Self-reflection, using video, has been described as a powerful technique for promoting change in adult behavior (Fukkink, 2008) and we may have seen greater teacher improvement if video reflection also had been used in TEEM.

4.2. Effect of the interventions on children's outcomes

4.2.1. Cognitive skills

In contrast to our hypotheses, there were no significant TEEM main effects on children's language and literacy skills. But there was a significant PALS main effect for children's print knowledge, demonstrating that children whose parent received the PALS intervention showed greater gains in their understanding of letters and letter-sound relations compared to children whose parents did not receive PALS. This finding suggests that PALS parents were providing their children with experiences that supported certain early literacy skills independent of their children's classroom designation. Although the PALS intervention did not explicitly target parents' attending to letters and print, the PALS curriculum videos did show examples of parents interacting with their children during play with puzzles, books, and alphabet-related toys (e.g., alphabet blocks, shape sorters) and these clips often showed parents talking with their children about letters and the sounds they made. The parents in the PALS intervention may have found these types of interactions helpful as a model for playing with their children in similar ways.

There was also an unexpected TEEM by PALS interaction showing that the children in the TEEM classrooms whose parents did not receive PALS had greater gains in phonological awareness. Because we expected the TEEM program to result in early literacy gains, it is difficult to explain why phonological skills improved more for children in TEEM without PALS. Overall, the few effects of TEEM on children's cognitive outcomes are contrary to expectations and difficult to interpret, given the greater gains seen in responsive behaviors and instructional practices for TEEM as compared to control teachers. A possible explanation for these null findings is that the duration of the teacher intervention was insufficient. In another study examining the effectiveness of TEEM that included Head Start, child care, and public school pre-k teachers, we found significant positive effects for teachers after the first year of the intervention, but effects on children's outcomes were not seen until the second year (Landry, Smith et al., 2011; Landry, Swank et al., 2011). This finding suggested that teachers were not able to implement new skills to the extent required to impact children's outcomes during the initial year. It was not until they had "made these skills their own" in the second year that positive effects on children's learning were observed.

It may also be that the increases seen for the TEEM teachers' instructional practices were simply not large enough to facilitate greater increases in children's learning. Teachers in both groups had pretest scores that were at the low-moderate end of the TBRS scale, and even though the TEEM teachers made greater gains than the control teachers, by midyear and posttest their scores were not yet at the 3.0 level on a 4-point rating scale (see Table 2). Previously, we found that ratings in instructional areas on the TBRS need to move to a moderate-high level of 2.8 or greater level in order to see significant increases in child outcomes (Landry, Swank, Smith, Assel, & Gunnewig, 2006). Thus, steeper gains across the school year may be required to impact children's learning within the same school year, which may be difficult when teachers start at such low levels of instruction and only receive one year of intervention.

Another possible explanation is that instruction was not rigorous enough to move children's cognitive outcomes. As children gain in their social and cognitive abilities, teachers must introduce

the instruction of more complex pre-literacy skills and language (e.g., blending and elision, letter-sound relations, inferential questioning). Although some teachers make this shift from basic to advanced instructional targets easily, other teachers have more difficulty, and thus, many children may plateau in their learning. More attention during the midyear period to the professional development of teachers who have difficulty introducing instruction that builds more complex skills may be needed to achieve greater child gains in school readiness. Additionally, designing studies so that teachers who need this degree of support can have two years of training would allow for the evaluation of whether the gains necessary to see strong outcomes for children can be achieved with longer periods of professional development support.

4.2.2. Social skills

Results for reports of children's social competence and behavior were comparable in the TEEM and control classrooms. However, children in the TEEM classrooms showed significantly greater decreases in teacher-reported school avoidance as compared to children in the control classrooms ($d = -0.24$), and a noteworthy effect for greater gains in school liking ($d = 0.23$), suggesting that increases in the TEEM teachers' responsive behaviors and best practices supported children's desire to attend school. Although, this is a teacher-report measure, studies have shown good agreement between independent observations and teacher-report measures (Howes & Ritchie, 1999).

4.2.3. Self-regulation

The greatest gains in self-regulation were seen for children whose parents received the PALS intervention, as compared to children whose parents did not receive PALS, regardless of their TEEM status ($d = 0.19$), although one TEEM by PALS interaction was noteworthy ($d = 0.18$). The task battery used in this study included both "hot" (affectively salient) as well as "cool" (affectively neutral) self-regulation tasks. Interestingly, PALS was found to improve children's hot, but not cool self-regulation, consistent with one prior study of a PALS-based intervention conducted in family child care homes (Merz, Landry, Johnson, Williams, & Jung, 2016). Specifically, on the gift delay-wrap task, children whose parents received PALS were better able to withhold the urge to peek while the gift box was being wrapped. Based on the attachment theoretical framework, warm acceptance of the child's interests with responses that are prompt and contingently linked to what the child signals facilitates the child's regulation of stress and novelty, which allows the child to trust his/her needs will be met when complying with behavioral limits. Thus, improving parental responsiveness may have strengthened the child's sense of security in his/her relationship with the parent, reducing stress and allowing independent self-regulatory skills to develop.

Consistent with the PALS finding for the gift delay-wrap task, children whose parents received PALS improved in parent-rated (but not teacher-rated) effortful control compared to those whose parents did not receive PALS ($d = 0.15$). The parent-rated effortful control composite was an average of the CBQ Attention focusing and Inhibitory Control scales. Parent rating scale measures of self-regulation have often been considered to be more reflective of hot than cool self-regulation. One possible explanation for why there were effects for the parent- but not teacher-report measure is that children who experienced more responsive parenting demonstrated greater effortful control in the home but not school.

Providing PALS to parents when their children are preschool-age takes advantage of a critically important period during early childhood when self-regulation skills are rapidly developing (Blair, 2002; Garon et al., 2008). By improving children's early self-regulatory development, PALS could potentially prevent later academic and social problems, as children who are dysregulated

fare poorly in social interactions with peers and are more likely to doubt their ability as good learners (Raver, Garner, & Smith-Donald, 2007). Development of behavioral control in emotionally charged situations may be better supported by parents in the home environment, versus in the classroom, as there are more frequent opportunities for one-on-one responsive interactions tailored to the child's signals and needs. For example, a recent intervention in Head Start classrooms to support hot and cool self-regulation found significant gains in cool self-regulation but not hot self-regulation. The authors suggested that future studies may need to attend to parent processes in addition to classroom practices to see effects on hot self-regulation (Raver et al., 2011).

4.2.4. Children's behaviors during book reading and free play

Contextually relevant observational measures of mother-child interactions were also used to evaluate increases in school readiness skills, as standardized assessments often fall short in capturing children's behaviors related to functioning in everyday contexts. We felt that measurement of mother-child interactions was important to include because of the role of social interaction in the development of school readiness skills, particularly self-regulation (Hala, Pexman, Climie, Rostad, & Glenwright, 2010). For example, the shared book reading and toy play contexts placed demands on children to use skills related to communication and self-regulation, such as forming appropriate responses to requests and developing ideas and sharing them through initiations directed to their mother. To accomplish this, children are required to identify goals, sequence their behavior to carry out goals, and respond flexibly to feedback, behaviors important for the development of self-regulation. Positive effects in favor of children whose parents received PALS and who were in a TEEM classroom were seen for positive engagement during shared book reading, a rating that required a complex set of behaviors including focused attention, initiating interactions, and enthusiastic involvement. This TEEM by PALS finding suggests that children's experiences in their TEEM classroom contributed to increases in their social interaction skills with their parent, in addition to their parent's increased use of responsive behaviors as a result of participation in PALS. Although we saw only one significant TEEM by PALS interaction for gains in children's interactive behaviors, there were notable effect sizes for shared enjoyment ($d = 0.32$) and initiative ($d = 0.42$) that suggest an added benefit of the combination of the TEEM and the PALS intervention over and above either one of them alone. Observations of the videotaped book reading activity also showed that children whose parents received PALS achieved greater gains in their use of language, including asking questions, making comments, and engaging in discussion about the book with complete sentences. While we did not see PALS effects for the standardized language measures, the observed interaction shows that PALS did impact children's use of language in an activity with a familiar person, suggesting that observational measures may add information about which aspects of child language can be facilitated by different types of interventions.

Finally, more than one third of participating families reported speaking Spanish at home, and 17% reported speaking a combination of English and Spanish. Analyses of the child language measures administered in Spanish (to a qualifying sub-sample of 222 Spanish-speaking children) showed no significant group differences and small effect sizes (median effect size = 0.02), suggesting that this was not due to a problem of low power due to a small sample size. However, as most of the children received their instruction in English in the classroom, this might be expected. It is possible that children who were learning both English and Spanish may have been impacted differently by the interventions. The effects of parent interventions on children's language growth may depend in part on congruence between the home and school languages (Cummins, 1980; Dressler & Kamil, 2006). It may also take time

for increases in responsive parenting to translate into benefits for children's linguistic development in their second language.

4.3. Limitations and future directions

Although we made ongoing efforts to keep families in the study until completion, our attrition rate of 30% is one limitation of this study. Our attrition analyses established that there were no significant differences in the patterns of missingness between groups, suggesting that our results were not biased by the attrition. Nonetheless, if a sizable percentage of parents do not complete an intervention (or control condition), we may need to consider modifications that would enhance retention, such as shortening the intervention, increasing incentives for completion, or perhaps offering the intervention in a more convenient format such as an online version (for those in the intervention condition). It should be noted, however, that our attrition rate is lower than other large home visiting studies, which are often reported to have attrition as high as 50% (e.g., 57% in McGuigan, Katzev, & Pratt, 2003; 36% in Fernandez & Eyberg, 2009 57% in Wagner, Cameto, & Gerlach-Downie, 1996). Families from low-SES backgrounds such as those in Head Start programs have many life challenges (e.g., multiple jobs, illness, frequent moves, etc.) that make it difficult for them to stay engaged in a program. Also, the Head Start agencies that we worked with had strict rules about school attendance, and a number of children were dropped from their school program due to poor attendance, or parents sought other programs (e.g., public school pre-k).

A second limitation may have been a lack of explicit attention to fostering an enhanced home-school connection as part of the intervention. While our interventions in both environments were aligned in many ways, it is possible that there was still not enough direct linkages between home and school activities to achieve the desired interactive effect of our dual intervention model. Future efforts might benefit from adding a component to both the PALS and TEEM interventions focused on building closely aligned home-school activities.

5. Conclusions

Similar to recent shared book reading studies (Justice et al., 2015; Lonigan & Whitehurst, 1998), the hypothesis that simultaneous, aligned teacher-plus-parent interventions would be more powerful than the classroom program alone or the home program alone was not broadly supported. The only instance of a significant, positive interaction was for children's engagement during book reading for students experiencing both the classroom and home programs. Findings of this study demonstrate the important value of a responsive parenting intervention for preschool-aged children's development of a range of school readiness skills. While Head Start programs value the involvement of parents, they often do not incorporate a parent program with proven efficacy (Cooper & Lanza, 2014). In light of findings from the Head Start Impact study (USDHHS & ACF, 2010) showing that a classroom experience alone is not enough to ensure that children from low-SES homes will enter school ready to succeed, a greater commitment to including an effective parent program could change this picture. This study also suggests that teachers may require a more sustained intervention to adopt instructional practices of high enough quality to, in turn, impact child outcomes. Although our theory of change was only partially supported, future research should continue to examine potentially positive effects of intervening with aligned approaches in both the school and home environments.

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Appendix A. Supplementary data

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